Deependra Singh

deependra.singh@emory.edu | deependra230.github.io

Academic Employment

Emory University — Atlanta, GA
Visiting Assistant Professor, *Department of Mathematics* — Aug 2024 - Present

Education

University of Pennsylvania — Philadelphia, PA
Ph.D., Mathematics — 2019 – 2024
Advisor: David Harbater

University of Pennsylvania — Philadelphia, PA
M.S.E., Computer & Information Science (GPA 4.0/4.0) — 2017 – 2019

Indian Institute of Technology Delhi — New Delhi, India
B.Tech., Computer Science & Engineering — 2010 – 2014

Publications & Preprints

- Local–global principles for spin groups over semi-global fields, with V. Suresh. *In preparation.*
- Admissible groups over number fields. *Journal of Algebra* 690 (2026), 832–859.
- Machine learning approaches to the Shafarevich–Tate group of elliptic curves, with A. Babei, B. S. Banwait, AJ Fong, and X. Huang. *Advances in Theoretical and Mathematical Physics* 29(8) (2025), 2353–2379.
- Enhanced Integrated Gradients: improving interpretability of deep learning models using splicing codes as a case study, with A. Jha, J. K. Aicher, M. R. Gazzara, and Y. Barash. *Genome Biology* 21 (2020), Article 149.
- Automating relevance banding in e-Commerce search using Click Model, with V. Deolalikar. *IEEE International Conference on Big Data* (2018), 3793–3797.
- Islands of Interest: Mining concentrations of user search intent over e-commerce product categories, with N. Pradhan and V. Deolalikar. *IEEE International Conference on Big Data* (2018), 3717–3722.

Teaching

Emory University — Atlanta, GA

MATH 221: Linear Algebra (2 sections)	Spring 2027 (scheduled)
MATH 211: Multivariable Calculus (2 sections)	Fall 2026
MATH 221: Linear Algebra (2 sections)	Spring 2026
MATH 211: Multivariable Calculus	Fall 2025
MATH 112: Calculus II	Fall 2025
MATH 211: Multivariable Calculus (2 sections)	Spring 2025
MATH 221: Linear Algebra (2 sections)	Fall 2024

University of Pennsylvania — Philadelphia, PA
Instructor, MATH 1410: Multivariable Calculus — Summer 2023

Penn Pre-First Year Program — Philadelphia, PA
Program for first-generation, low-income undergraduate students.

Instructor, MATH C: Pre-Calculus	Summer 2024
Teaching Assistant, MATH A: Pre-Calculus	Summer 2023

Princeton Prison Teaching Initiative — New Jersey
Volunteer math instructor for South Woods State Prison.

Tutor, MATH 020: Algebra	Spring 2024
Co-Instructor, Math in the Courtroom	Spring 2023

University of Pennsylvania — Philadelphia, PA
Teaching Assistant

MATH 3610: Advanced Calculus II	Spring 2024
MATH 360: Advanced Calculus I	Fall 2021
MATH 509: Honors Analysis II	Spring 2021
MATH 240: Linear Algebra and ODEs	Fall 2020

Undergraduate Advising

Emory University — Atlanta, GA
REU Project Mentor — Summer 2026
Project: *AI-Assisted Exploration in Algebra and Number Theory*
Students: Taylor Murrell and Casey Simon

Penn Directed Reading Program — Philadelphia, PA
Mentored the following undergraduate reading projects:

Elementary Number Theory, Molly Bradley	Spring 2024
Quadratic Reciprocity, Angela Cai	Fall 2023
Algebraic Curves, Max Wang	Spring 2023
Probabilistic Methods, Max Wang	Fall 2022
Covering Spaces, Kevin You	Fall 2021

Awards and Fellowships

Certificate in College and University Teaching — Fall 2023
Penn Center for Teaching and Learning

Good Teaching Award — Spring 2021
University of Pennsylvania Mathematics Department

Benjamin Franklin Fellowship — 2019–2024
University of Pennsylvania

Fellowship in Computer Science — 2017–2019
University of Pennsylvania

MHRD Scholarship — 2010–2014
Government of India

Academic Service and Outreach

Organizer, Emory Algebra and Number Theory Seminar	Fall 2024–Present
Co-organizer, Joint Athens–Atlanta Number Theory Seminar	Fall 2025
Co-organizer, Emory Learning Seminar on Stacks and Brauer Groups	Spring 2025
Co-organizer, Emory Learning Seminar on Patching over Semi-global Fields	Fall 2024
Co-organizer, Emory Learning Seminar on Étale Cohomology	Fall 2024
Co-organizer, Penn Graduate Algebraic Geometry Seminar	Fall 2022–Spring 2024
Co-organizer, Patching in Arithmetic Geometry Seminar	Spring 2022
Co-organizer, Math Circles, West Philadelphia High School	Summer 2023

Talks

Research Talks

Arithmetic Geometry Seminar, Ohio State University	Spring 2026
Algebra and Number Theory Seminar, Emory University	Fall 2024
Graduate Conference in Algebra, Geometry, and Topology, Temple University	Summer 2024
Algebra Seminar, University of Pennsylvania	Spring 2024

Selected Expository Talks

Emory Learning Seminar on Stacks and Brauer Groups — Spring 2025
Twisted sheaves and Gabber's theorem on Brauer groups
Several talks based on M. Olsson's book *Algebraic Spaces and Stacks*

Emory Learning Seminar on Étale Cohomology — Fall 2024
Several talks based on J. Milne's book *Étale Cohomology*

Penn Graduate Algebraic Geometry Seminar

Rational points on varieties and the Brauer–Manin set	Spring 2024
An introduction to étale cohomology	Spring 2023
Admissibility over $\mathbb{Q}$	Fall 2022

Patching in Arithmetic Geometry

Patching division algebras over semi-global fields	Spring 2022

Other Expository Talks

Grothendieck topologies and cohomology	Spring 2024
Milnor K-theory	Spring 2022
Numerical invariants and stability on algebraic stacks	Fall 2021
Introduction to spectral sequences	Spring 2021

Industry Employment

Groupon — Palo Alto, CA
Research Scientist, *Search & Data Mining team* — Oct 2015 – June 2017

Groupon — Chennai, India
Data Scientist, *Data Science team* — Jun 2014 – Sep 2015